APS106

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more while loops.

Week 4 | Lecture 2 (4.2)

Upcoming

- Lab 2 Due 11:59 pm Friday.
- No labs released this week.
- Reflection 4 Released Friday 6:00 pm.
- Tutorial (Online), Practical, Office Hour sessions running all week.

if nothing else, write **#cleancode**.

This Week's Content

- **Lecture 4.1**
  - function review, while loops
- **Lecture 4.2**
  - **more loops**
- **Lecture 4.3**
  - Objects & Strings: Operators and Methods

While Loops

- The **while loop** keeps executing a piece of code as long as a particular condition is **True**.
- There must be a colon (:) at the end of the while statement.
- The action to be performed must be indented.

Must evaluate to True or False

Colon

```
while expression:
    do something.
```

Indent

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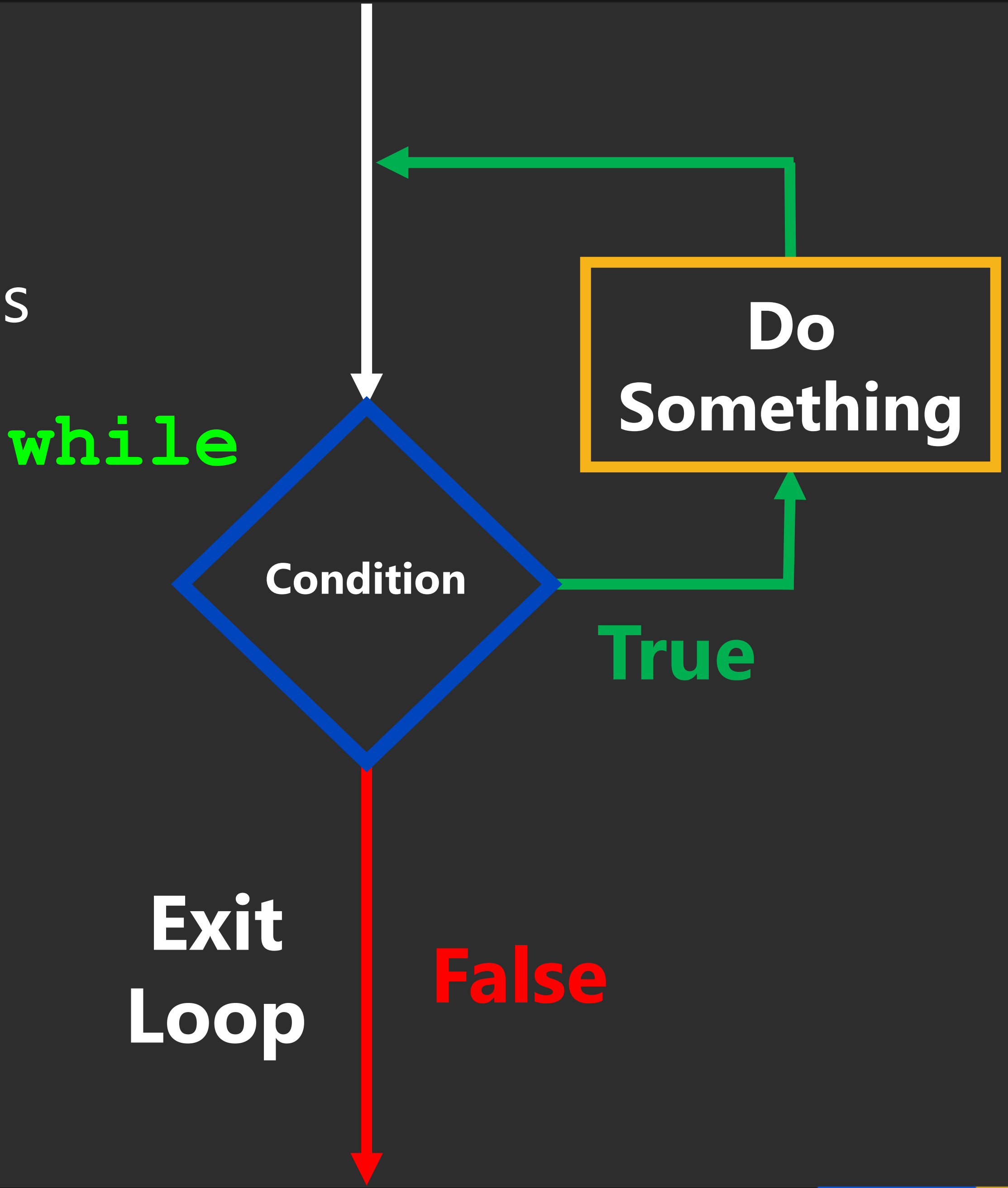
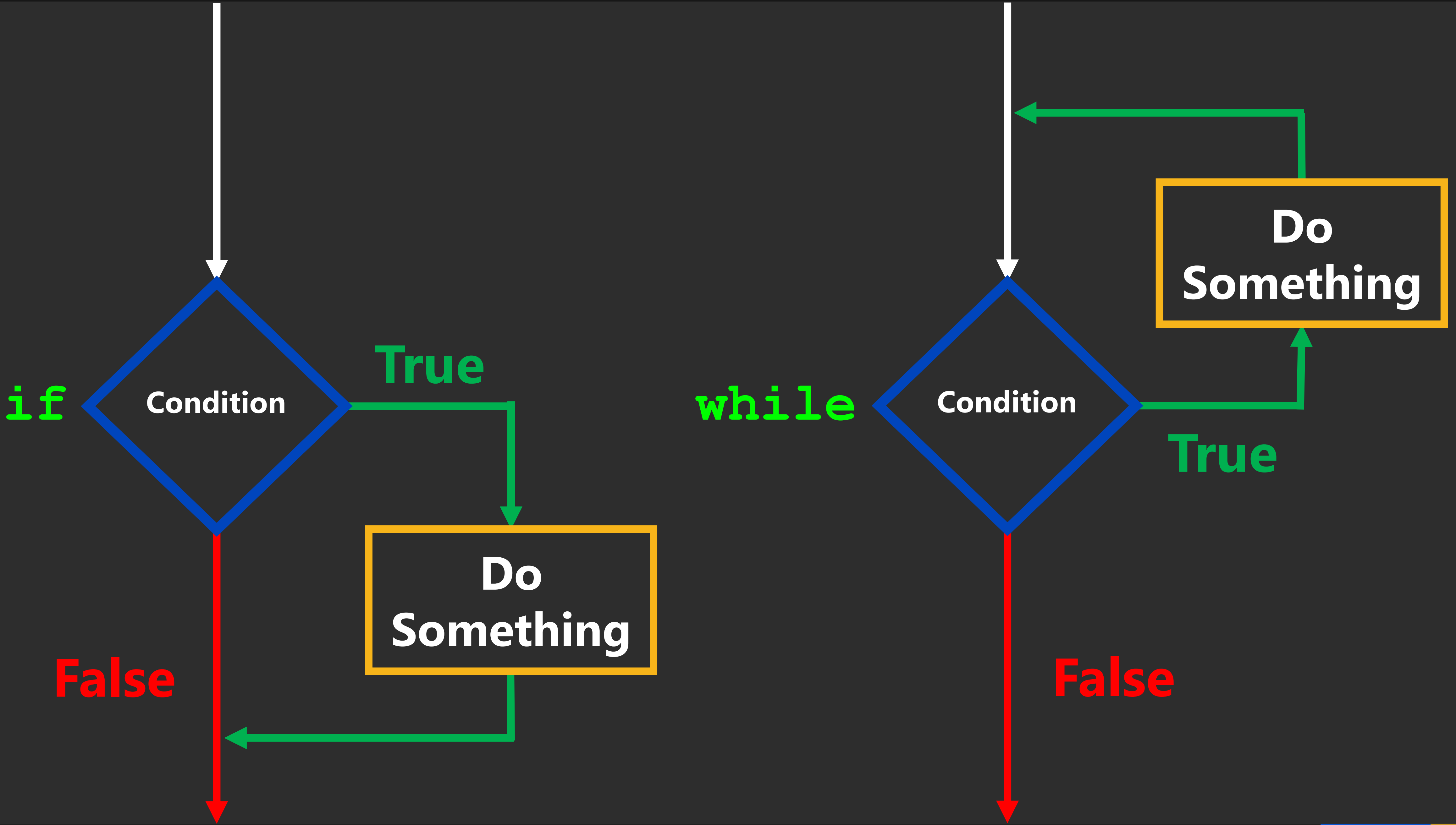
While Loops

- The condition that gets evaluated is just an boolean expression.
- In particular it can include:
  - Something that evaluates to **True** or **False**.
  - logical operators (**and**, **or**, **not**)
  - comparison operators
  - function calls
- ... really anything that evaluates to **True** or **False**.




if
Condition
True
Do Something
False
while
Condition
True
Do Something
False

Refresher

- How many printouts will the following **while** loop produce?

```
x = 1
while x < 4:
    print(x)
    x = x + 1
```

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Click Link:

1. Refresher

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Refresher

- Just like for **if**-statements, if you use **and** or **or** in a while-loop expression, it is subject to lazy evaluation.
- Only if **x < 4** is **True** will **y < 4** be evaluated. #solazy

```
while x < 4 and y < 4:
      ...
```

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Click Link:
2. Lazy Evaluation

Random Module

- This module implements pseudo-random number generators for various distributions.

```
import random

random.uniform()
random.random()
random.randint()
...
```

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Click Link:
3. Random Module

Breakout Session 1

- Write a function that roles a 6-sided dice until a lucky number is rolled.

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Click Link:
4. Breakout Session 2

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Breakout Session 2

- Write some code to allow someone to play Rock, Paper, Scissors, Lizard, Spock repeatedly until they beat the computer `3` times.

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Click Link:

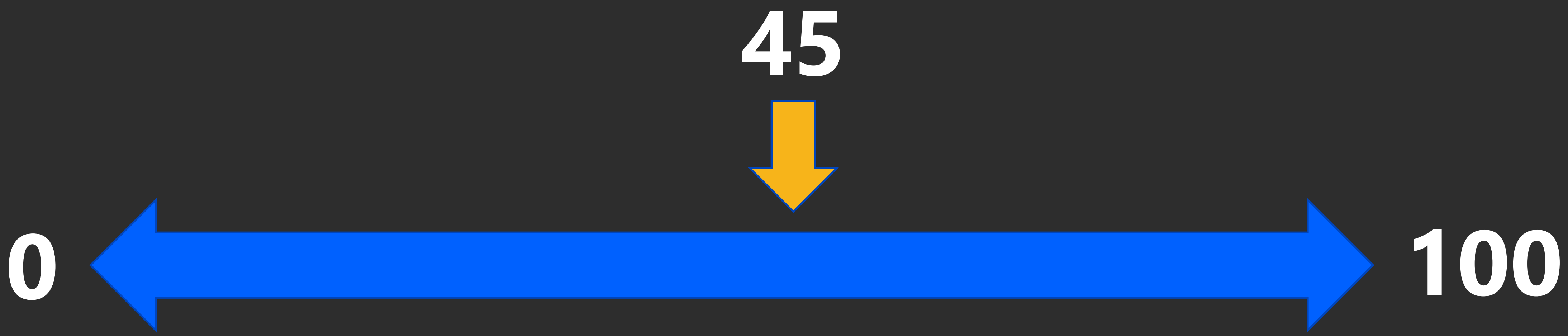
5. Breakout Session 1

Guessing Game

- Let's build a simple guessing game.
    - Get the computer to choose a random integer from 0 to 100.
    - Ask the user for a guess and allow the user to input a guess or "q".
    - If the user inputs "q" print a nice message and end the program.
    - If the user enters a guess, tell them if they should guess higher, lower, or if they got it right.
    - If they got it right, print a nice message and quit.

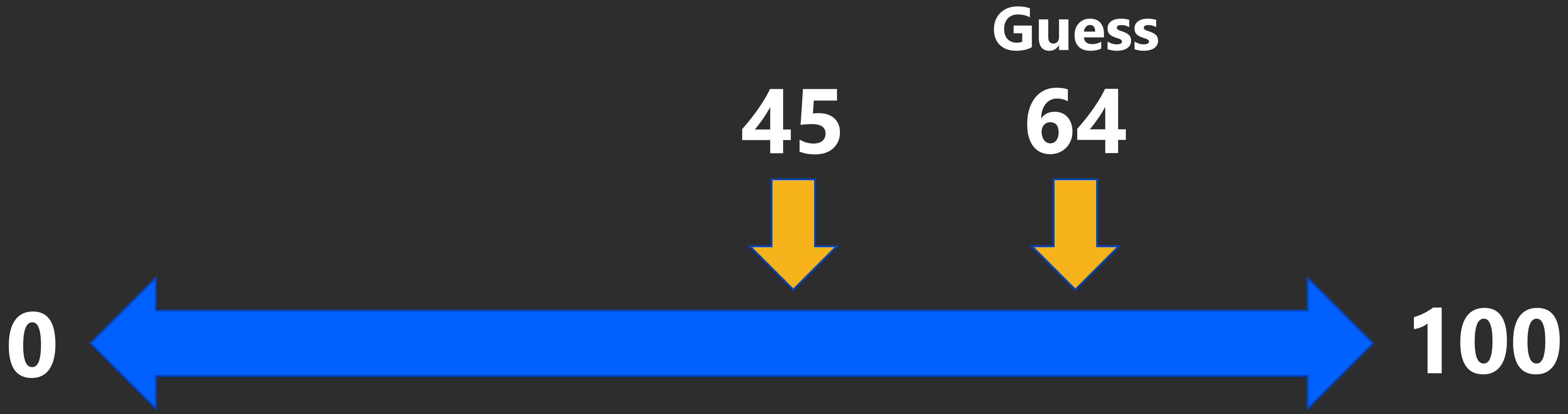
Guessing Game

- Get the computer to choose a random integer from 0 to 100.
  - The computer selects 45.

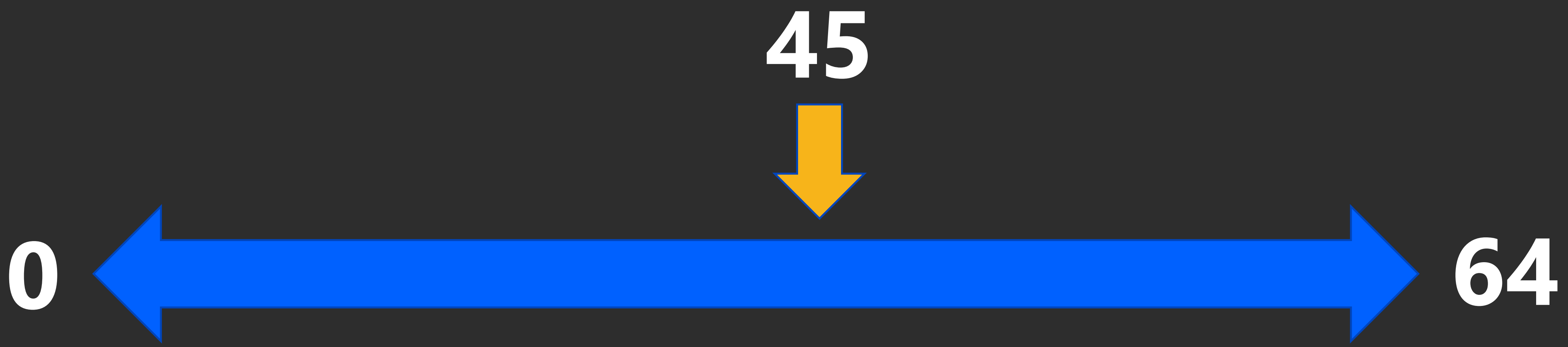


Guessing Game

- The user guesses 64.
  - The computer says **LOWER**.

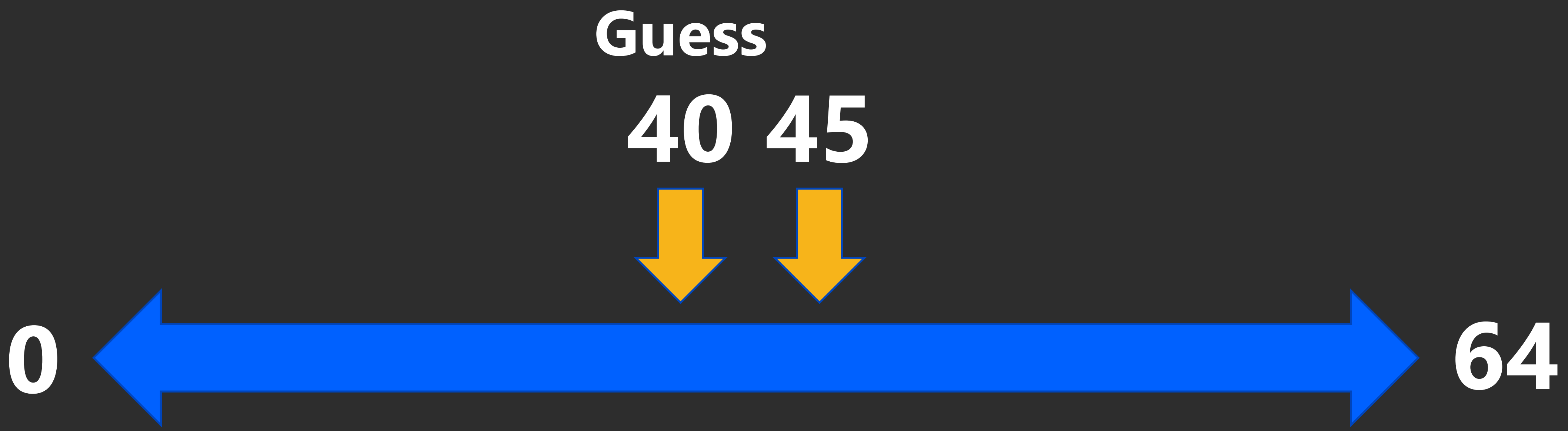


Guessing Game

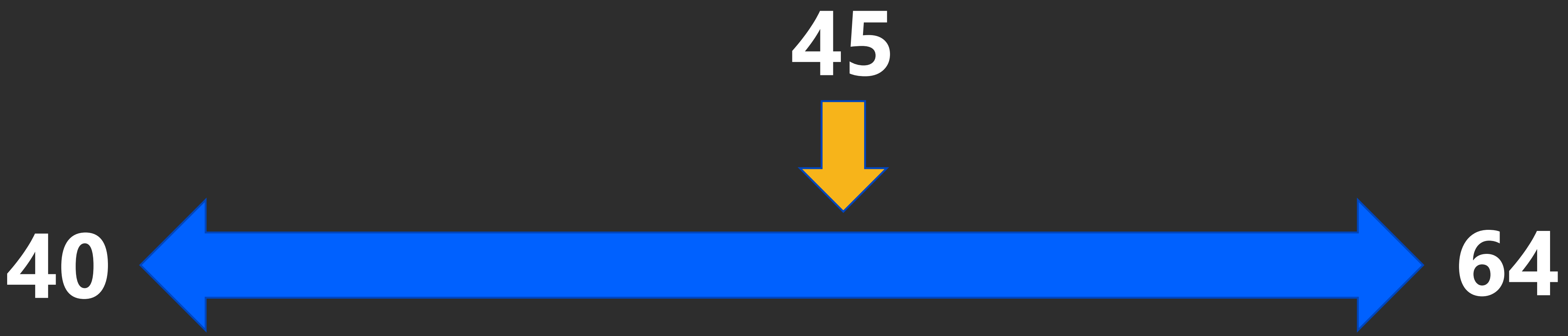


Guessing Game

- The user guesses 40.
  - The computer says **HIGHER**.

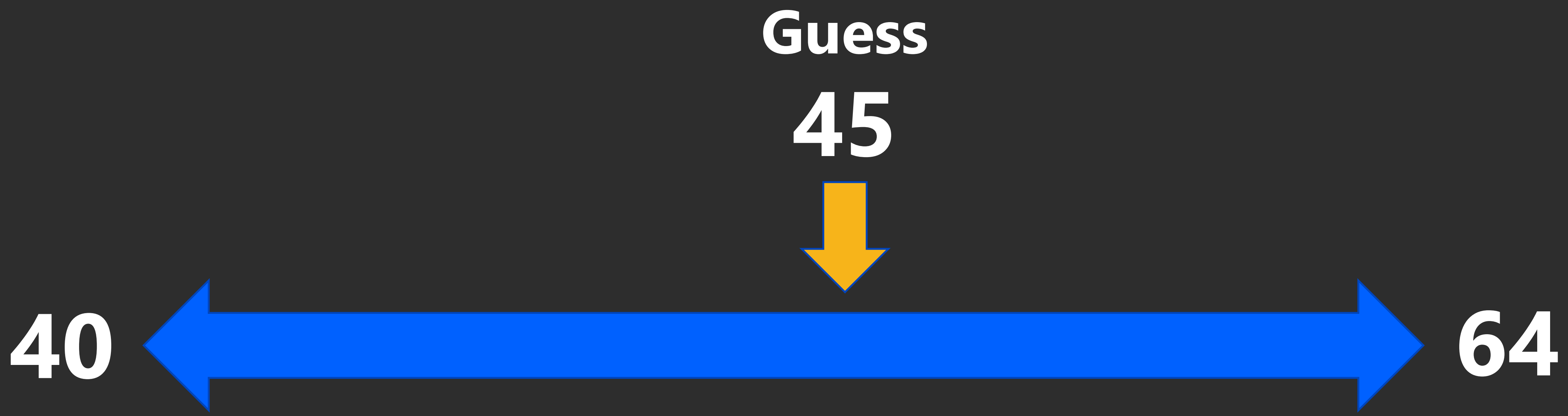


Guessing Game



Guessing Game

- The user guesses 45.
  - The computer says **YOU WIN**.



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Guessing Game

- Let's build a simple guessing game.
  1. Get the computer to choose a random integer from 0 to 100.
  2. Ask the user for a guess and allow the user to input a guess or "q".
  3. If the user inputs "q" print a nice message and end the program.
  4. If the user enters a guess, tell them if they should guess higher, lower, or if they got it right.
  5. If they got it right, print a nice message and quit.

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Click Link:
6. A Simple Guessing Game

PRACTICE!

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